

NLP Final Project: Qwen 2.5 Code Generation and LoRA Fine-Tuning Analysis

Student Name: Miray Balıkoğlu

1. Project Summary

In this study, the Qwen 2.5-Coder-1.5B-Instruct base model was fine-tuned on two different datasets, "Deep" and "Diverse," using the LoRA (Low-Rank Adaptation) technique, and its Python code generation performance was analyzed.

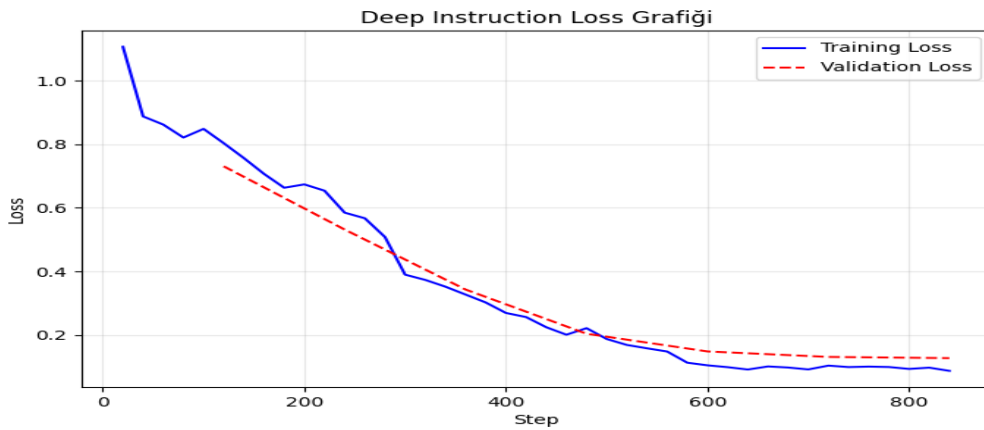
2. Technical Configuration

Training was conducted on an A100 GPU; logs were recorded every 20 steps, and validation every 100 steps. Epoch=3, Batch Size=4, and Context Length=1024 were used for both models. The following parameters were customized according to the nature of the datasets:

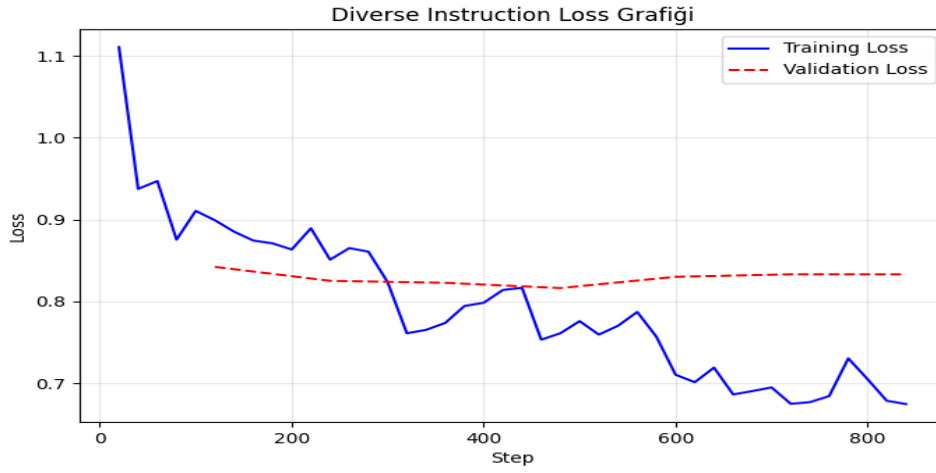
- **Deep Instruction Model:** Using standard configuration, Rank (r)=32, Alpha=64, Dropout=0.1, and Learning Rate=2e-4 were used.
- **Diverse Instruction Model:** Capacity was increased due to data complexity; Rank (r)=64, Alpha=128, and Dropout=0.15 were selected. For stabilization, the Learning Rate was reduced to 1.5e-4.

3. Training Process and Loss Analysis

When examining the error rates (Loss) during the training process, two different behaviors were observed:



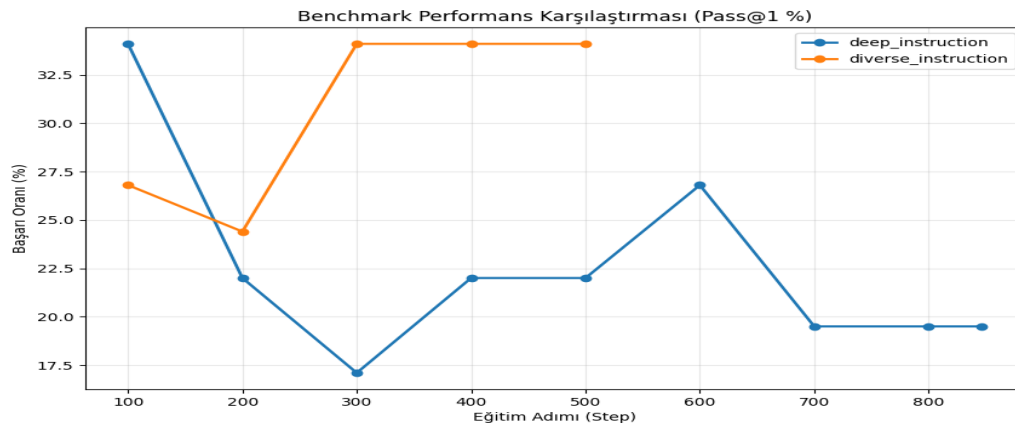
- **Deep Model:** Training Loss decreased steadily from 0.96 to 0.19. Validation Loss decreased until step 600 and then plateaued, remaining stable.



- **Diverse Model:** Although Training Loss continued to decrease, Validation Loss began to rise again after step 500 (0.43 -> 0.47). This clearly indicates that the model lost its generalization ability after step 500 and started memorizing the data (Overfitting).

4. Benchmark Performance Analysis (Checkpoint Comparison)

The success (Pass@1) of checkpoints taken during training on 41 questions from the AtCoder (Easy) dataset is compared in the graph below.



Analysis on the graph:

- **Deep Model (Blue Line):** Reached its highest success (34.1%) at the very beginning of training (Step 100), but performance dropped dramatically as training progressed. This shows that while the model overly adapted to the format in the "Deep" dataset, it rapidly lost its general coding ability.

- **Diverse Model (Orange Line):** Exhibited a more stable graph. It reached its highest performance (34.1%) between Step 300 and Step 500 and maintained this level for a while. However, consistent with the overfitting signals in the Loss graph, improvement ceased towards the end of training.

Results and Evaluation

As a result of the tests performed, the best scores of the Base Model and Fine-Tune models are as follows:

Model	Best Checkpoint	Pass@1 (%)	Solved / Total
Base Model	-	41.46%	17 / 41
Deep Model	step-100	34.1%	14 / 41
Diverse Model	step-300	34.1%	14 / 41

General Evaluation:

1. **Catastrophic Forgetting:** Although the fine-tuning process enabled the models to learn the specific instruction format, it caused a decline in the Base Model's general problem-solving ability (41.4% -> 34.1%).
2. **Impact of Diverse Dataset:** The Diverse model exhibited a slower and more stable learning curve compared to the Deep model (the plateau of the orange line in the graph), but experienced overfitting in the final steps due to the high Rank (64) selection.
3. **Suggestion:** In future studies, to preserve Base Model performance, the Learning Rate should be further reduced, and examples from the original dataset (Replay Buffer) should be added to the training data.

DEEP-DIVERSE COMPARİSİON

#	Soru ID	Deep Instruction (14 Doğru)	Diverse Instruction (14 Doğru)	Durum
1	abc365_a	✓ PASS	✓ PASS	Ortak Başarı
2	abc370_a	✓ PASS	✓ PASS	Ortak Başarı
3	abc372_a	✓ PASS	✓ PASS	Ortak Başarı
4	abc373_a	✓ PASS	✓ PASS	Ortak Başarı

#	Soru ID	Deep Instruction (14 Doğru)	Diverse Instruction (14 Doğru)	Durum
5	abc374_a	✓ PASS	✓ PASS	Ortak Başarı
6	abc380_a	✓ PASS	✓ PASS	Ortak Başarı
7	abc384_a	✓ PASS	✓ PASS	Ortak Başarı
8	abc378_a	✓ PASS	✗ FAIL	Sadece Deep
9	abc366_a	✓ PASS (Ek)	✗ FAIL	Sadece Deep
10	abc367_a	✓ PASS (Ek)	✗ FAIL	Sadece Deep
11	abc369_a	✓ PASS (Ek)	✗ FAIL	Sadece Deep
12	abc371_a	✓ PASS (Ek)	✗ FAIL	Sadece Deep
13	abc375_a	✓ PASS (Ek)	✗ FAIL	Sadece Deep
14	abc382_a	✓ PASS (Ek)	✗ FAIL	Sadece Deep
15	abc368_a	✗ FAIL	✓ PASS	Sadece Diverse
16	abc368_b	✗ FAIL	✓ PASS	Sadece Diverse
17	abc374_b	✗ FAIL	✓ PASS	Sadece Diverse
18	abc377_b	✗ FAIL	✓ PASS	Sadece Diverse
19	abc377_a	✗ FAIL	✓ PASS	Sadece Diverse
20	abc379_b	✗ FAIL	✓ PASS	Sadece Diverse
21	abc379_a	✗ FAIL	✓ PASS	Sadece Diverse

□ Diverse

Daha geniş bir veri çeşitliliği ile eğitildiği için standart soru kalıplarını "ezberleme" yeteneği yüksektir. Basit ve doğrudan kodlama gerektiren sorularda, soruyu aşırı analiz etmeden (over-thinking yapmadan) en kestirme yoldan çözüme ulaşmıştır.

□ Deep

Adım adım düşünme" yeteneği sayesinde, sorulardaki gizli detayları ve istisnai durumları (edge-cases) daha iyi yakalamıştır. Diverse modelin yüzeysel okuyup hata yaptığı mantık ağırlıklı sorularda, derinlemesine analiz yaparak doğru sonucu bulmuştur.

Ortak Başarının Nedeni

Her iki modelin de aynı soruları çözebilmesinin temel nedeni, bu soruların **düşük karmaşıklıkta ve net** olmasıdır.

- **Temel Yeterlilik:** Ortak çözülen sorular, genellikle temel sözdizimi (syntax) ve basit mantık gerektiren problemlerdir. Her iki model de temel kodlama yeteneklerine sahip olduğu için bu standart sorularda zorlanmamıştır.
- **Yöntem Farkının Önemsizleşmesi:** Sorular karmaşık uç durumlar (edge-cases) veya dolambaçlı ifadeler içermediği için, Deep modelin "derin düşünme" yeteneğine veya Diverse modelin "çeşitlilik" avantajına ihtiyaç duyulmamış; temel bilgi yeterli olmuştur.